

Research Papers

A study on nano-graphene oxide surface modification for the design of paraffin/graphene oxide phase change material

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ABSTRACT

Incorporation of surface modified nanoparticles into paraffin can produce a phase change material (PCM) with superior thermal conductivity and better heat storage capacity. In the present research, for the first time, graphene oxide (GO) nanoparticles were modified with ethoxytrimethylsilane (ETMS) to improve their dispersion and stability in paraffin-based PCMs. Composite PCMs with different amounts of unmodified and modified GO (MGO) nanoparticles (0.25 wt%, 0.5 wt%, and 1 wt%) were prepared and evaluated with various analyses including X-ray diffraction (XRD), Fourier transform infrared spectroscopy (FTIR), differential scanning calorimetry (DSC), atomic force microscopy (AFM), thermal conductivity measurement, thermogravimetric analysis (TGA), viscosity test, and melt fraction measurement. FTIR results confirmed the successful decoration of ETMS on the surface of GO nanoparticles. AFM observation corroborated that the MGO nanoparticles have an average thickness of 1–2 nm, which is almost similar to that of a single layer GO. DSC results demonstrated that compared to pure paraffin, the melting enthalpy decreases by 6–10 % with the inclusion of unmodified nanoparticles, while it increases by 0–8 % with the inclusion of modified nanoparticles. TGA results verified that the thermal stability in nanocomposites is improved compared to pure paraffin, and this improvement is more significant for the samples with modified nanoparticles. That initial degradation temperature increased about 13–17 °C and 22–26 °C for nanocomposites with GO and MGO nanoparticles, respectively. Rheological study confirmed that the viscosity and modulus of nanocomposites with MGO were much higher than those of pure paraffin and nanocomposites with GO nanoparticles and the loading of 0.25 % MGO leads to the greatest increase in viscosity and modulus. Repeated viscosity measurement disclosed that the viscosity stability of nanocomposites with MGO is higher than the nanocomposites with GO. Thermal conductivity measurement revealed that thermal conductivity in nanocomposites is enhanced compared to pure paraffin by 13–39 %, and this enhancement is more evident for the samples with unmodified nanoparticles. Melting fraction measurement proved that the melting rate in nanocomposites is higher than pure paraffin and the fastest melting is obtained for the sample with 0.25 % nanoparticles. The time required for complete melting was reduced by 12–25 % in nanocomposite with 0.25 % GO compared to the pure paraffin.

1. Introduction

Energy is stored as sensible heat, latent heat, thermochemical heat or a combination of them. A latent heat storage system by phase change materials (PCMs) is much more advantageous than other methods because it has high energy density and can be performed at a constant temperature [1]. PCMs are classified into different groups, including

organic, inorganic, and eutectic materials. It is worth mentioning that each of these groups generally does not have the expected properties on their own, and their properties and performance should be improved by using different methods [2].

In the group of organic PCMs, there are paraffinic and non-paraffinic components. Paraffinic components are formed from saturated hydrocarbons and are considered the most widely used PCMs. The properties

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Fig. 1. Photograph of synthesized nanographene oxide.

of different paraffinic components are almost similar and this makes their study easier [2]. Kabbel and Abdelgaied [3] used paraffin as phase change material in solar energy storage and found that paraffin as a determining factor can increase the efficiency of solar energy utilization by 67 %. Although paraffin is one of the most common phase change materials due to its high enthalpy, its low thermal conductivity reduces

the rate of heat transfer and the efficiency of energy storage. It has been proved that the low conductivity of paraffin can be increased by adding nanoparticles [4,5], so that in recent years, nanofluids prepared by dispersing nanomaterials in the base phase have received much attention for energy storage applications [6–8]. In addition to low thermal conductivity, paraffin faces the problem of leakage during phase change. Compositing with nanomaterials can also solve this problem.

Graphene oxide is a layered carbon nanomaterial that consists of hydrophilic oxygenated graphene sheets with many functional groups. Like other carbon nanomaterials, graphene oxide possesses a very high

Table 1
Instruction of synthesis of P/GO and P/MGO nanocomposites.

Sample code	Sample name	Paraffin (%)	GO (%)	MGO (%)
P	Paraffin	100	0	0
0.25GO	Nanocomposite	99.75	0.25	0
0.5GO	Nanocomposite	99.5	0.5	0
1GO	Nanocomposite	99	1	0
0.25MGO	Nanocomposite	99.75	0	0.25
0.5MGO	Nanocomposite	99.5	0	0.5
1MGO	Nanocomposite	99	0	1

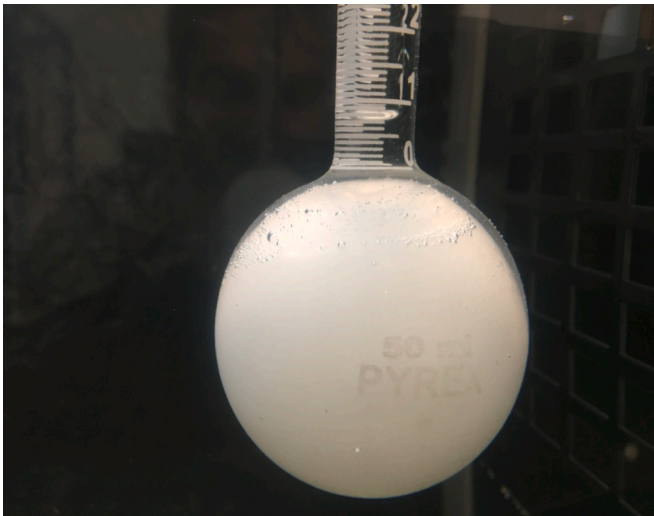


Fig. 3. Photograph of melting nanocomposite.

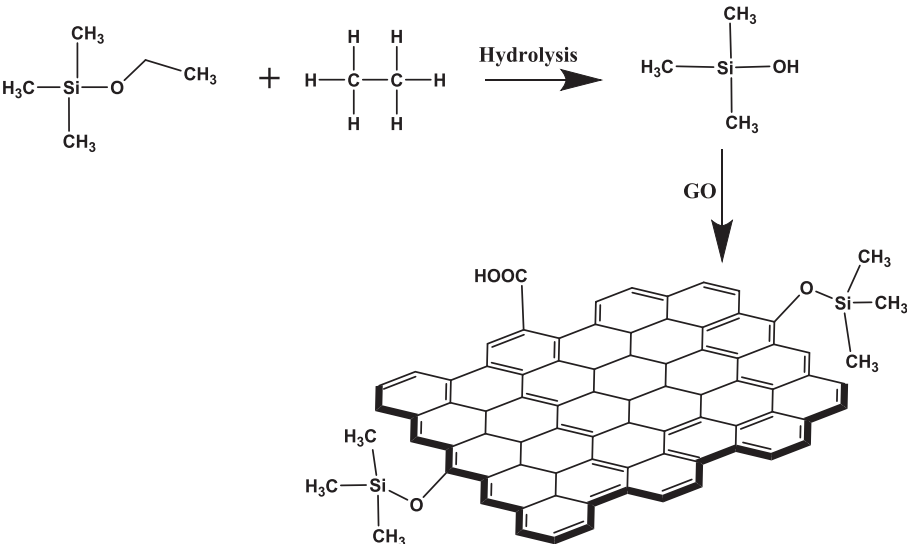


Fig. 2. Schematic of surface modification reaction of GO.

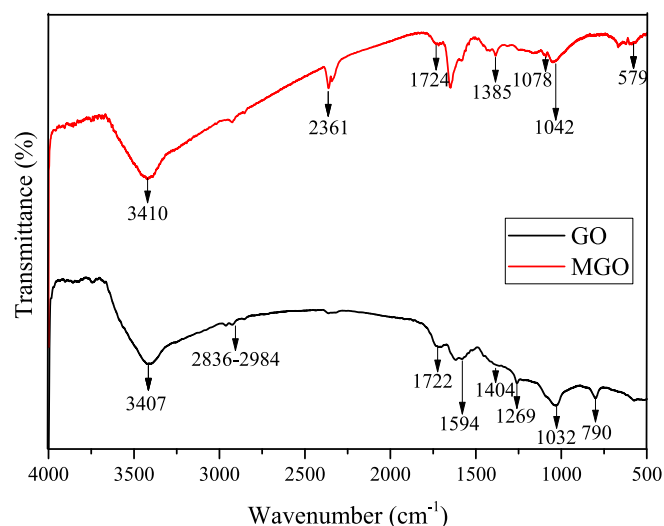


Fig. 4. FTIR spectra of GO and MGO nanoparticles.

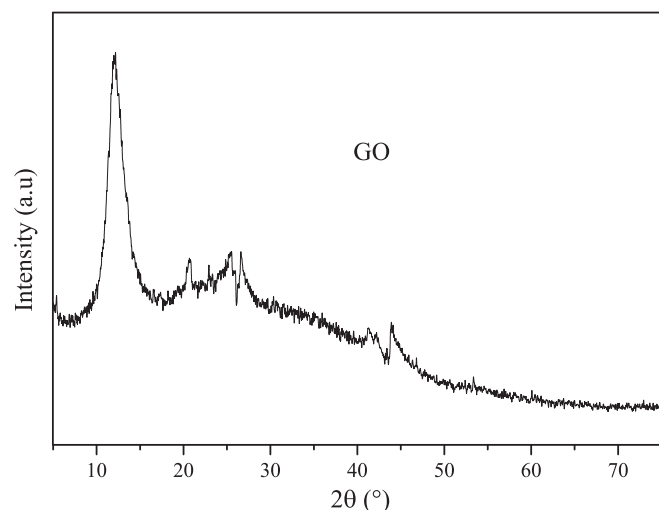


Fig. 5. XRD pattern of GO.

thermal conductivity coefficient. Also, the presence of functional groups on graphene oxide has made it able to form composites with other materials. Moreover, these functional groups provide appropriate sites for further surface modification and decoration of various chemical moieties on nanoparticle surface [9–11].

It has been documented that the incorporation of graphene and graphene oxide into paraffin can significantly increase the thermal conductivity of the resulting PCM. For instance, Zhao et al. [12] confirmed that the addition of 0.2 wt% graphene oxide to the paraffin emulsion increases the thermal conductivity by >100 %. Similarly, Xiang et al. [13] reported that the incorporation of 1.2 wt% graphene into the paraffin emulsion raises the thermal conductivity by >20 %. Chen et al. [14] declared that incorporating 0.03 wt% graphene oxide into the paraffin emulsion enhances thermal conductivity by about 10 %.

Graphene, graphene oxide and their derivatives have been extensively used for fabrication of phase change materials [15–19]. In this regard, Barison et al. [20] added graphene oxide and reduced graphene oxide nanoparticles to a paraffin-based nanoemulsion and confirmed that both nanoparticles increase thermal conductivity, improve emulsion stability, and decrease subcooling of the PCM. They declared that the effect of GO and rGO on the thermal properties of the PCM was

almost the same, however, rGO increased the optical properties to a higher extent and resulted in better solar absorption ability.

In another research, Zheng et al. [21], employed graphene oxide and copper foam to improve thermal conductivity and shape stability of the paraffin. To prepare the composites, graphene oxide was reduced in the presence of ascorbic acid, then the copper foam was submerged in the reduced graphene oxide solution and freeze-dried, the resulting 3D structure was immersed in molten paraffin, and paraffin was combined with the structure via vacuum impregnation approach. As a result of this procedure, a PCM composite with almost 65 wt% paraffin, 5 wt% graphene oxide, and 30 wt% copper foam was achieved. The thermal conductivity, shape stability, and solar to thermal energy conversion of the composite were substantially higher than those of pristine paraffin. Especially, the thermal conductivity of the composite was augmented by 1000 % compared to pristine paraffin.

Gao et al. [22], employed reduced graphene oxide in addition to carbonized melamine foam to boost the thermal and stability properties of the paraffin. For this purpose, the melamine foam was immersed in the rGO suspension, dried in an oven, and then carbonized at high temperature. The carbonized matrix was then loaded with liquid paraffin under vacuum. The tri-component composites with varying amounts of rGO were developed. Thermal conductivity and thermal stability as well as shape and cyclic stability were improved in the composites compared to pristine paraffin. The higher the amount of rGO, the higher the thermal conductivity and the better the stability of the composite.

He et al. [23] evaluated the effect of reduction extent of graphene oxide on energy storage and thermal behavior of paraffin-based composites. Graphene oxide was reduced through a hydrothermal process with different temperatures, including 95, 120, and 400 °C. Research results corroborated that with the increase of temperature, the reduction extent was raised leading to improved thermal conductivity and increased photothermal conversion efficiency. Moreover, the paraffin leakage was substantially decreased and the shape stability was markedly promoted in accordance with the increase of reduction extent. Because with the increase of reduction extent, the number of oxygen bearing groups and structural defects of the graphene oxide decreased.

Chen et al. [14] loaded different amounts of graphene oxide including 0.01, 0.02, and 0.03 wt% in paraffin emulsion and revealed that the thermal conductivity increases proportionally with the amount of nanoparticles. However, 0.02 wt% of graphene oxide was selected as the optimum amount, because 0.03 wt% of graphene oxide substantially increased the viscosity of the composite and consequently decreased the flowability of the composite.

Although graphene and graphene oxide are considered as suitable nanomaterials to improve the thermal properties and energy storage ability of PCMs, their improper dispersion in matrix is probable, which leads to the formation of structural defects in the composite and the decrease of the anticipated performance. To surmount this issue, surface modification of these nanoparticles has been paid attention. Surface modification can not only improve the dispersion state of nanoparticles, but also augment some other qualities like photothermal conversion ability of the composite [24].

In this regard, Selvaraj and Krishnan [25] loaded surface functionalized graphene nanoplatelets into PCM to enhance its performance for thermal management applications. Graphene nanoplatelets were treated with a mixture of nitric acid and sulfuric acids and then loaded into a polymer matrix with varying volume ratios in the range of 0 to 0.3 vol%. Acid treatment created a large number of functional groups, including COOH, OH, and C=O on the graphene surface and improved its interaction with the polymer matrix. The inclusion of functionalized graphene nanoplatelets in the polymer matrix augmented its thermal conductivity and thermal diffusivity. The higher the graphene loading, the higher the thermal conductivity and thermal diffusivity. Moreover, the stability of nanocomposites under repeated heating was higher than pristine polymer. More importantly, the specific heat capacity was

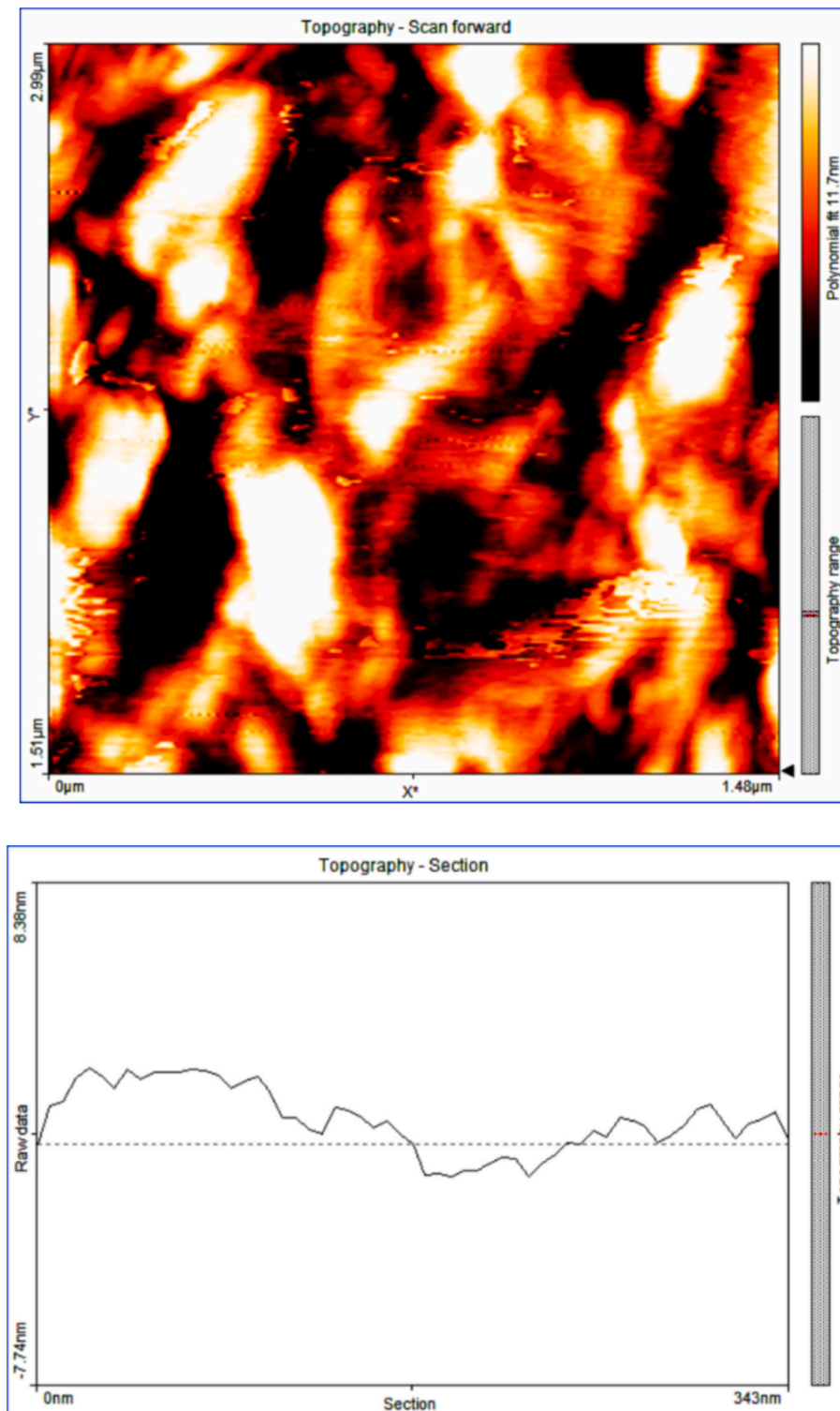


Fig. 6. AFM images of GO. AFM images taken in tapping mode and line profiles of exfoliated graphene oxide nano sheets deposited from dispersions in water onto mica substrate (for image, height profile taken along the indicated lines are presented).

raised by incorporating functionalized graphene nanoplatelets up to 20 vol%, most likely due to the confinement effect of nanoplatelets on phonon vibrations.

In another research, Nagar et al. [26] added carboxyl functionalized graphene nanoparticles to paraffin to enhance its heat storage ability. Nanocomposites with various ratios of carboxyl functionalized graphene ranging from 0.25 to 1 vol% were prepared and characterized. The characterization results showed that thermal conductivity, thermal

diffusivity and energy storage ability increase proportionally with the graphene ratio.

According to what was mentioned above, graphene oxide has been widely used to improve the thermal conductivity and heat storage ability of paraffin-based PCMs. Moreover, the influence of surface modification and the creation of different functional groups on graphene oxide on dispersion of nanoparticles in paraffin and the energy storage performance of the resulting composites has been extensively evaluated.

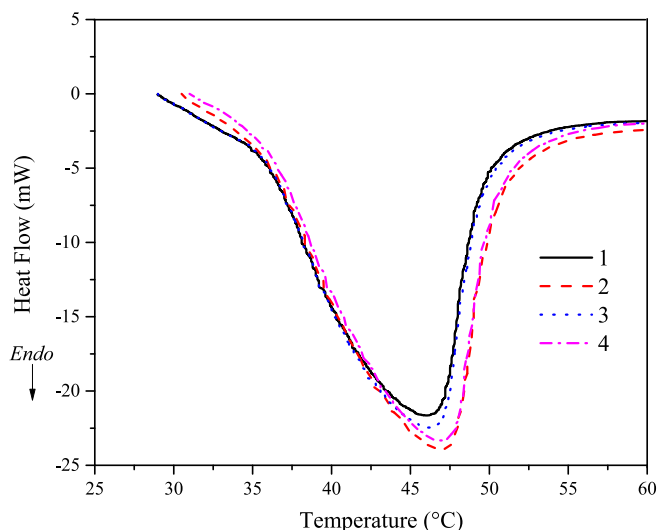


Fig. 7. DSC curve of paraffin, with four repetitions.

Although, a large number of paraffin-based PCMs loaded with unmodified and surface-modified graphene oxide nanoparticles have been prepared and investigated so far, a composite containing ethoxytrimethylsilane-modified graphene oxide nanoparticles has not yet been fabricated. The decoration of ethoxytrimethylsilane groups on

the surface of graphene oxide can strongly prevent nanoparticle agglomeration due to the substantial steric hindrance. Moreover, surface modified nanoparticles make proper interactions with paraffin due to having methyl and ethoxy groups, which leads to good dispersion of nanoparticles throughout the composite and the formation of a 3D network with heat transfer capability. In addition, the interaction between nanoparticles and paraffin as well as the formation of 3D network can prevent the leakage of paraffin during phase change.

Taking this into account, in the present research, it has been attempted to develop a composite by including graphene oxide modified with ethoxytrimethylsilane in paraffin to achieve a phase change material with improved thermal conductivity coefficient and superior energy storage ability. Different amounts of unmodified and modified GO nanoparticles including 0.25 wt%, 0.5 wt%, and 1 wt% were added to paraffin and the nanocomposites were evaluated with various analyses including X-ray diffraction (XRD), Fourier transform infrared spectroscopy (FTIR), differential scanning calorimetry (DSC), atomic force microscopy (AFM), thermal conductivity measurement, thermogravimetric analysis (TGA), viscosity test, and melt fraction measurement.

2. Experimental

2.1. Materials

Paraffin (RT35 grade) was obtained from Rubitherm, Germany. Ethoxytrimethylsilane (with a molecular weight of 118.25 g/mol and a

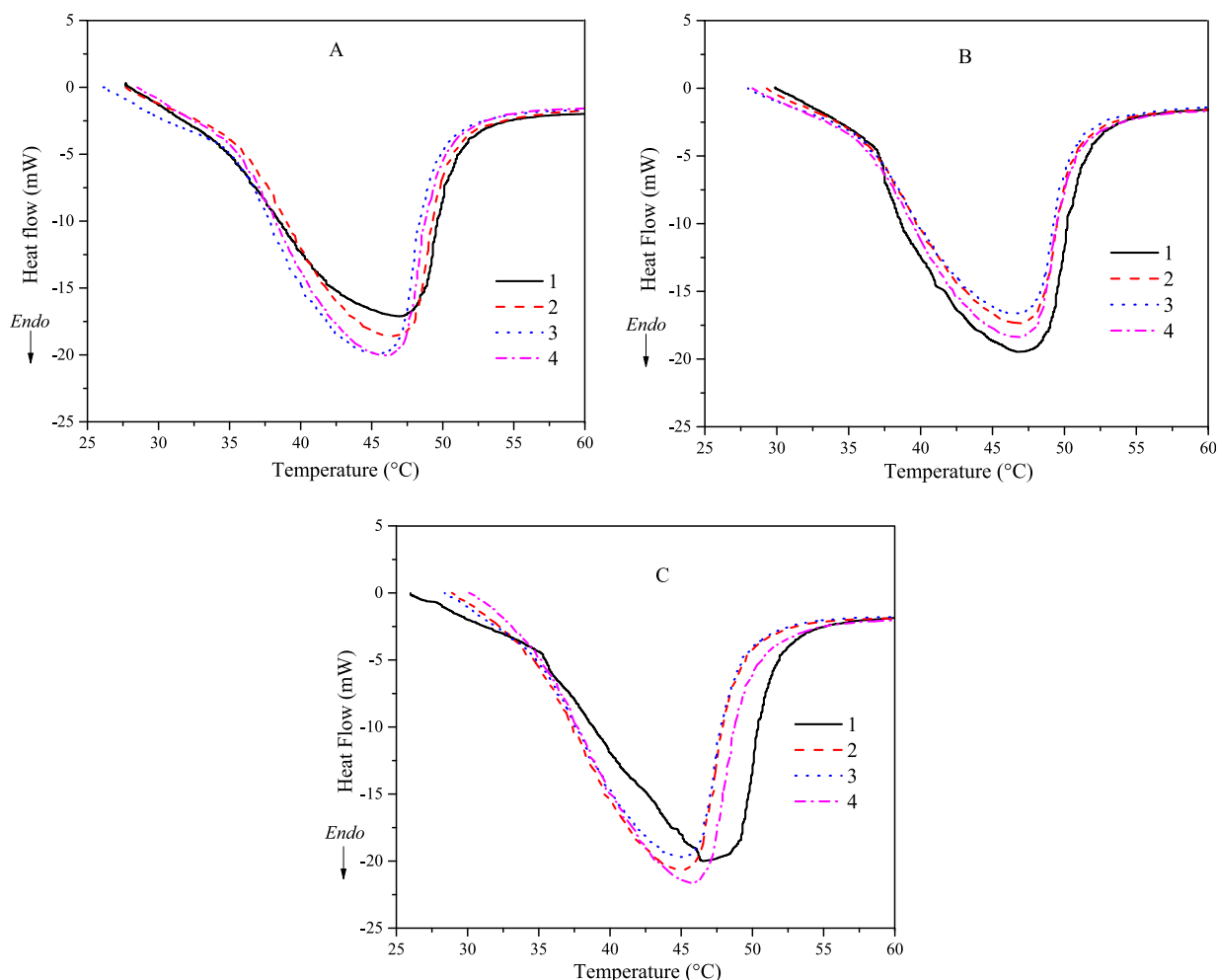


Fig. 8. DSC curve of nanocomposite P/GO, with four repetitions; A)0.25GO, B)0.5GO, C)1GO.

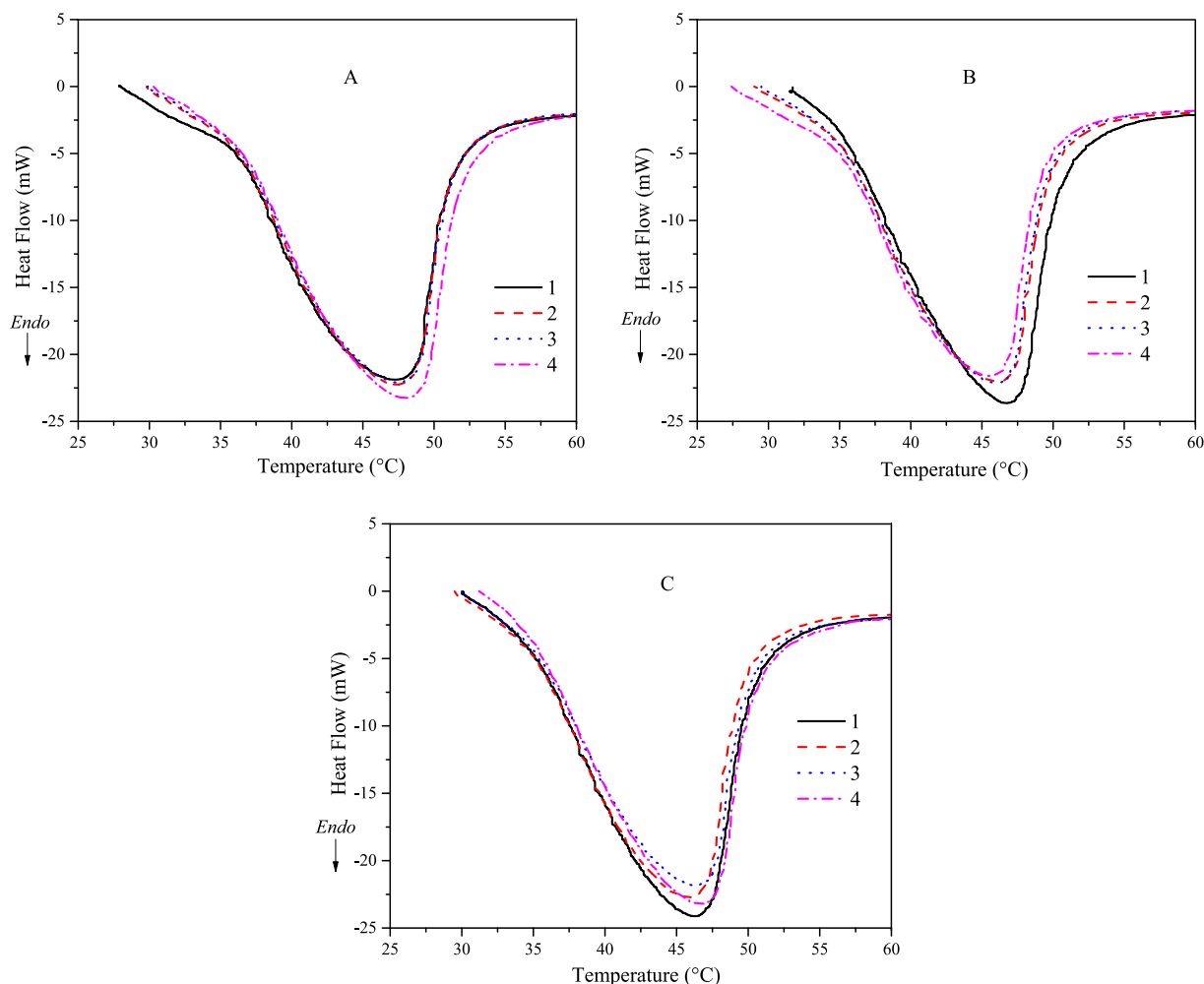


Fig. 9. DSC curve of nanocomposite P/MGO, with four repetitions; A) 0.25MGO, B) 0.5MGO, C) 1MGO.

density of 757 g/mol), sodium nitrate, and graphite for the synthesis of GO (12.01 g/mol molar ratio) were purchased from Sigma-Aldrich Company. Moreover, high-purity ethanol, sulfuric acid, methanol, hydrochloric acid, and hydrogen peroxide were obtained from the Luxury Chemical Industry Complex, and potassium permanganate was purchased from Scarlo Company.

2.2. Synthesis of nanographene oxide

To synthesize GO (Fig. 1), first, 1 g of graphite was mixed with 0.5 g of sodium nitrate. Then, 50 ml of sulfuric acid solution was added to the mixture and stirred for 30 min. Next, 7.3 g of potassium permanganate was gradually added to the mixture over 2 h, and the mixture was stirred at room temperature for 24 h. Finally, 50 ml of deionized water was added to the reaction mixture and stirred for 15 min. To terminate the reaction, 35 ml of hydrogen peroxide was added to the solution and stirred for 15 min. Then, the synthesized GO solution was washed three times with hydrochloric acid and distilled water, respectively. And at the end, the pH of the mixture was neutralized.

2.3. Modification of graphene oxide surface

To modify the surface of GO with ethoxytrimethylsilane, first, 10 mg of GO was dispersed in 50 ml of ethanol for 60 min, and then 5 ml of ethoxytrimethylsilane was added to the sample and the reaction mixture was stirred for at 60 °C 12 h. Finally, the sample was washed with methanol and placed in an oven to dry. In the GO surface modification

process, after adding silane, the solution reacts with the existing ethanol and the ethoxy group is separated from silane. From the same end, it reacts with OH branches on nanoparticles and bind to them. The schematic of this process is depicted in Fig. 2.

2.4. Preparation of samples

Paraffin was placed on a heater at a temperature of 50 °C until it was completely melted, then 0.25 wt%, 0.5 wt%, and 1 wt% of GO nanoparticles were added to it, and the samples were placed in a vibrating ultrasonic bath at 50 °C for 30 min. The nanocomposite was frozen and prepared for the test after it was probed with a power of 70 for 10 min. The percentage composition of the sample can be seen in Table 1.

2.5. Fourier-transform infrared spectroscopy (FT-IR)

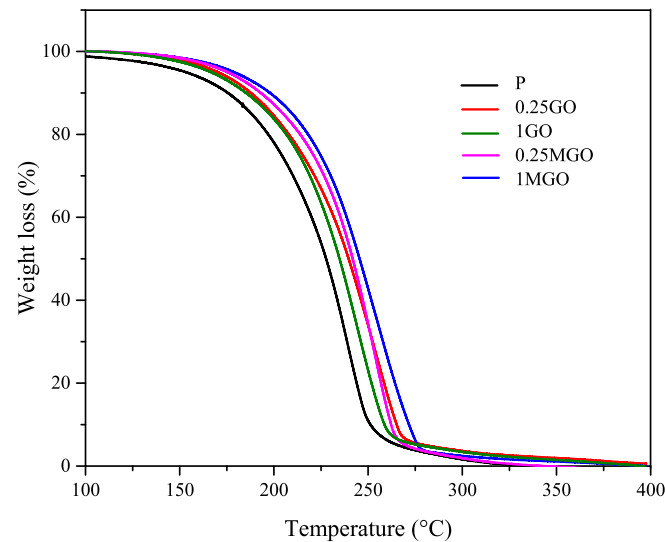
Fourier transform infrared spectroscopy (FTIR) was done with a Perkin-Elmer Spectrum GX model device in the wavelength range of 400–4000 cm^{-1} . FTIR was used to determine the structure of molecules, their structural similarity, and the type of functional groups and bonds.

2.6. Differential scanning calorimetry (DSC)

Differential scanning calorimetry was used to investigate the thermal behavior of materials. This operation was performed in the temperature range of 25 to 100 °C with a heating rate of 5 °C /min in a nitrogen environment.

Table 2
DSC curve data.

Sample name	Sample code	Melting point (°C)	Enthalpy J/g
P	R1	47.6	161.37
	R2	46.9	161.46
	R3	46.3	161.48
	R4	46.9	160.68
	Average	46.92 ± 0.53	161.25 ± 0.38
0.25 GO	R1	46.9	151.12
	R2	46.4	146.94
	R3	45.4	154.89
	R4	46	151.37
	Average	46.17 ± 0.63	151.08 ± 3.25
0.25 MGO	R1	47.3	172.8
	R2	47.4	176.92
	R3	47.5	175.39
	R4	47.9	174.5
	Average	47.52 ± 0.26	174.90 ± 1.72
0.50 GO	R1	46.8	146.32
	R2	46.8	141.55
	R3	46.5	146.86
	R4	46.6	153.27
	Average	46.67 ± 0.15	147.00 ± 4.81
0.50 MGO	R1	46.5	162.6
	R2	46.1	160.43
	R3	45.9	159.87
	R4	45.6	161.14
	Average	46.02 ± 0.38	161.01 ± 1.18
1.00 GO	R1	49	146.17
	R2	44.9	145.28
	R3	45	145.16
	R4	44.9	142.72
	Average	45.95 ± 2.03	144.83 ± 1.48
1.00 MGO	R1	47.4	172.96
	R2	46.3	173.98
	R3	45.9	172.6
	R4	46.3	166.35
	Average	46.47 ± 0.64	171.47 ± 3.46

**Fig. 10.** TGA thermograms of pure paraffin, P/GO, and P/MGO nanocomposites with 0.25 % and 1% nanoparticle concentrations.**Table 3**
TGA curve data.

Weight loss (%)	P	0.25GO	0.25MGO	1GO	1MGO
T _{5%}	153.34	170.27	175.35	166.57	178.86
T _{50%}	227.66	239.59	241.76	234.77	245.44
T _{90%}	251.22	262.17	265.15	258.65	271.68

Table 4
Thermal conductivity analysis data.

Sample	Thermal conductivity coefficient W/m.K
0.25GO (20 °C)	0.258
	0.254
	0.253
	0.248
	0.242
0.25GO (45 °C)	0.251 ± 0.006
	0.185
	0.145
	0.123
	0.122
0.25MGO (20 °C)	0.120
	0.139 ± 0.028
	0.229
	0.227
	0.226
0.25MGO (45 °C)	0.225
	0.223
	0.226 ± 0.002
	0.140
	0.139
	0.137
	0.137
	0.122
	0.135 ± 0.007

2.7. Atomic force microscopy (AFM)

Nano Wizard II model AFM device (JPK company, Germany) was used to conduct the imaging. Before imaging, a drop of diluted GO dispersion was poured on mica and dried at 45 °C

2.8. Thermal properties analyzer

The coefficient of thermal conductivity as the amount of heat energy that can pass through the material in a unit of thickness and a unit of time at a certain temperature was measured by the KD2 PRO device made by DECAGON, USA.

2.9. Thermogravimetric analysis (TGA)

Thermogravimetric analysis data were recorded with a Perkin Elmer STA6000 thermal analyzer. The samples were heated from ambient temperature to 600 °C at a rate of 10 °C/min.

2.10. X-ray powder diffraction analysis (XRD)

The XRD pattern of GO nanoparticles was analyzed with a D8-Advance diffractometer (Bruker company, Germany).

2.11. Viscosity test

In order to investigate how viscosity drops in thermal cycles, cooling-reheating has been performed. In this test, the stability and the drop of viscosity were checked at three constant temperatures with three repetitions. This test was done with Brookfield model DV2 machine made in America.

2.12. Rheology test (RMS)

The rheological measurement was performed using an MCR300 rheometer (with parallel plates, 25 mm in diameter, and 1 mm gap distance) manufactured by Paar, Germany. The temperature sweep test was performed with a frequency of 20 rad/s, a strain of 0.5 %, and a temperature range of 20 to 50 °C.

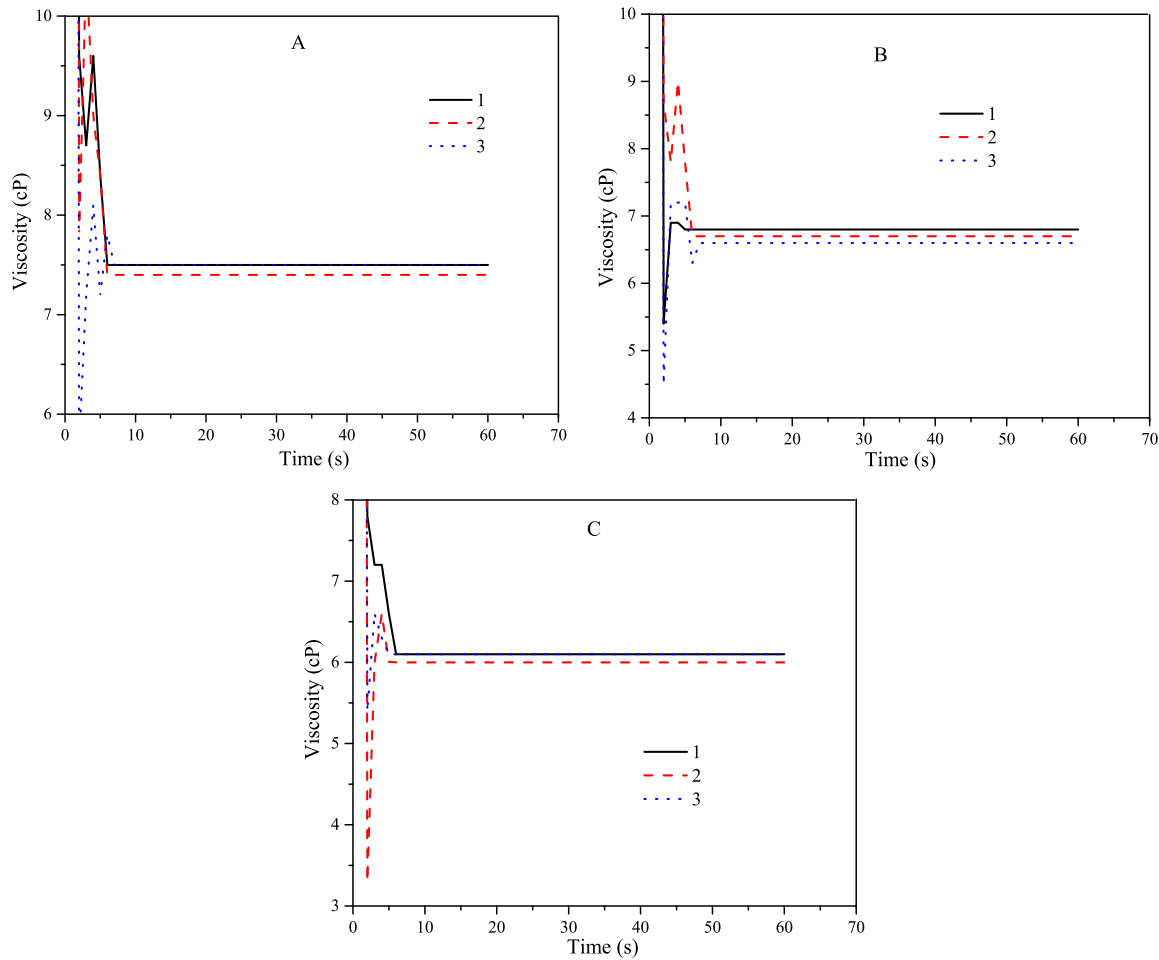


Fig. 11. Paraffin viscosity diagram. A) $T = 40\text{ }^{\circ}\text{C}$, B) $T = 45\text{ }^{\circ}\text{C}$, C) $T = 50\text{ }^{\circ}\text{C}$.

2.13. Melt fraction test

To perform this experiment, the required equipment mainly consisted of a spherical container, a water tank, a circulating water bath, and a thermostat chamber. The spherical container is made of graduated glass and its inner diameter is 1 cm (Fig. 3). In this experiment, paraffin and nanocomposites were tested by combining different percentages of nanoparticles at temperatures of 40, 45, and $50\text{ }^{\circ}\text{C}$. The melting fraction percentage is calculated by Eq. (1) [27]:

$$\frac{m}{M} = \frac{\rho_s \rho_l (V_c + V_e) - \rho_l M}{(\rho_s - \rho_l) M} \quad (1)$$

3. Results and discussion

3.1. FTIR results

The FTIR spectrum was used to confirm the synthesis of GO and the silanization reaction of GO nanoparticles. FTIR spectra of pristine graphene oxide and surface modified graphene oxide are represented in Fig. 4. In FTIR spectrum of GO, several characteristic peaks appeared. The peaks appearing at 3407 cm^{-1} , 1722 cm^{-1} , and 1404 cm^{-1} can be attributed to O—H stretching, C=O stretching, and O—H bending of oxygen-containing functional groups of GO surface. The peaks in the range of $2836\text{--}2984\text{ cm}^{-1}$ can be ascribed to C—H stretching of CH, CH₂, and CH₃ groups. The peak appearing at 1594 cm^{-1} can be attributed to C=C stretching of carbon ring. The peaks appearing at 1269 and 1032 cm^{-1} can be related to C—O stretching of epoxy groups and C—O stretching of alkoxy groups, respectively [27]. The peak appearing at

790 cm^{-1} can be assigned to C—H bending. The observed peaks were in line with those reported for graphene oxide in the literature [28,29].

In FTIR spectrum of MGO, in addition to the characteristic peaks of GO, new peaks also appeared. The peaks appearing at 2361 , 1385 , 1078 , 1042 cm^{-1} can be attributed to Si—C, Si—O—C, Si—O—Si, and Si—OH, respectively. Moreover, the peak appearing at 579 cm^{-1} can be related to Si—O—C bending [30–36]. The observed peak confirmed the successful modification of GO nanoparticles with ethoxytrimethylsilane.

3.2. XRD and AFM results

XRD spectrum of GO nanoparticles is shown in Fig. 5. In XRD spectrum, the main peak was detected at $2\theta = 12.16^{\circ}$ corresponding to the (002) plane. In the XRD spectrum of graphite, the main peak is usually observed at about 26° . In graphene oxide, the main peak was shifted to lower angle. The left-handed shift of the XRD peak corroborated that distance between the carbon layers in GO increases compared to graphite. This phenomenon is due to the formation of oxygen-containing groups on carbon rings and the weakening of van der Waals and π – π interactions between successive carbon layers, both of which prevent the stacking of carbon layers [27]. According to the Bragg equation, the smaller the Bragg angle, the higher the d-spacing. The d-spacing of GO nanoparticles was calculated by Bragg equation (Eq. (2)) and was found to be 0.73 nm .

$$n\lambda = 2d\sin\theta \quad (2)$$

Moreover, crystallite size of the nanoparticles calculated by Scherrer equation (Eq. (3)) was found to be 3.77 nm .

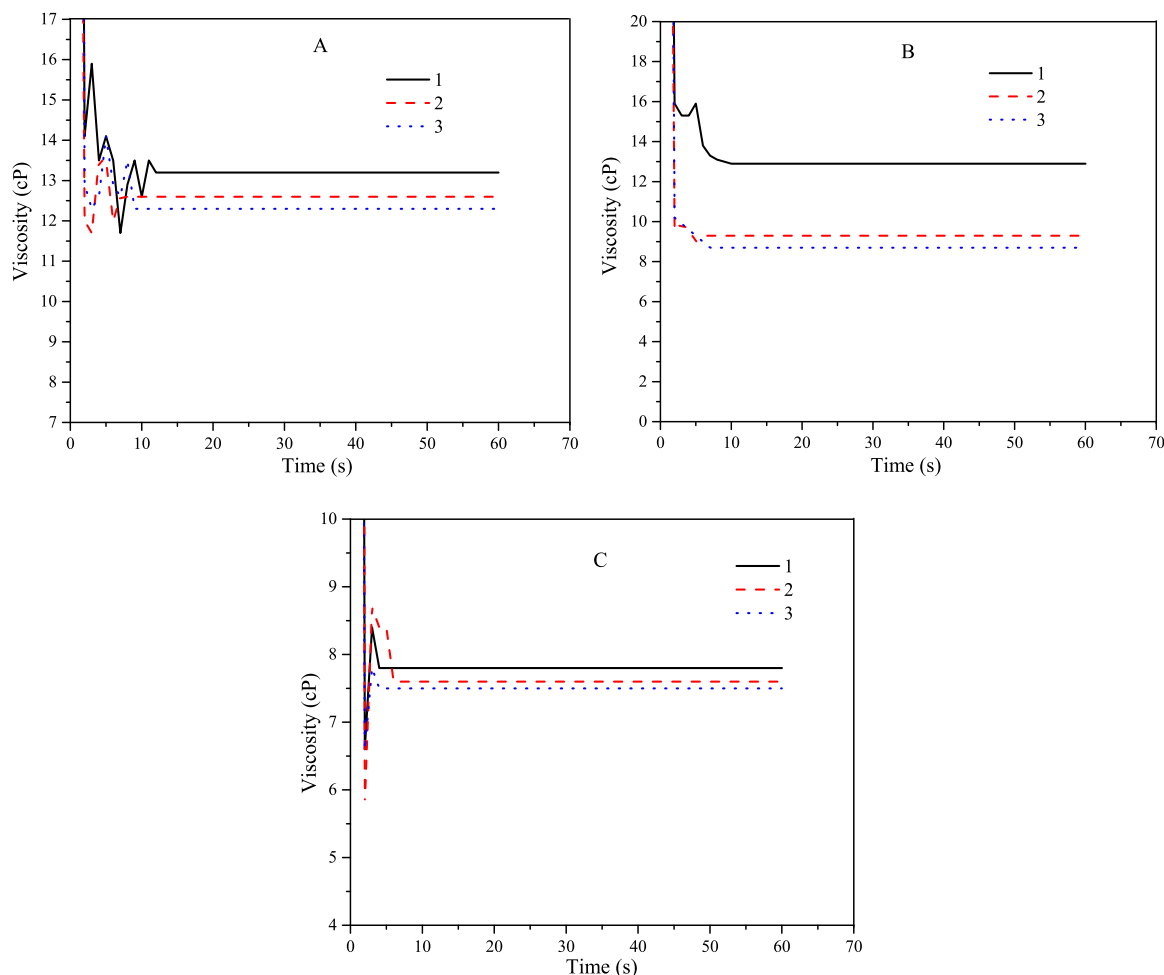


Fig. 12. P/0.25 GO nanocomposites viscosity diagram. A) $T = 40\text{ }^{\circ}\text{C}$, B) $T = 45\text{ }^{\circ}\text{C}$, C) $T = 50\text{ }^{\circ}\text{C}$.

$$D = \frac{K\lambda}{\beta \cos \theta} \quad (3)$$

where n , λ , d , θ , K , and β are diffraction order, X-ray wavelength (0.15418 nm), interplanar spacing, Bragg angle, shape factor, and full width at half maximum (FWHM) of a XRD peak, respectively. Other researchers also corroborated the left-handed shift of diffraction peak and the increase in d -spacing in GO compared to graphite and graphene [37,38].

Fig. 6 displays the AFM image and the corresponding height profile of GO sheets deposited on a mica substrate. According to AFM image, GO nanoparticles represented uniform thicknesses and different lateral dimensions. According to AFM image, the lateral dimensions of GO nanoparticles were in the range of 100 to 500 nm. According to the height profile, the thickness of GO nanoparticles was about 1–2 nm, which is somewhat higher than the values reported in literature for single layer GO most likely due to the formation of silane groups on the surface of GO.

3.3. DSC results

DSC was done to investigate the effect of GO nanoparticles on the thermal properties of paraffin. The thermal properties of the samples including melting temperature and latent heat capacity of melting were evaluated. DSC heating curves of pure paraffin as well as nanocomposites consisting of unmodified and modified GO are shown in Figs. 7, 8, and 9. Besides, the melting temperature and melting enthalpy of the samples are collected in Table 2. The main function of a PCM is to

store latent heat. In other words, PCMs store energy as the temperature rises and release it as the ambient temperature falls. Thus, the higher enthalpy of these materials, the higher the energy storage. Due to the low molecular weight and highly linear structure of paraffin, all its chains participate in crystallization and its crystallinity reaches about 100 %. When nanoparticles are added to paraffin, the enthalpy usually decreases because the nanoparticles prevent the complete crystallization of paraffin. However, in the case of PCMs, the enthalpy reduction is very small so that they can store energy well [39,40]. Fig. 7 depicts the DSC curve of paraffin with four repetitions. In DSC curves, an endothermic peak can be observed at about $46\text{ }^{\circ}\text{C}$ corresponding to the solid-liquid transition of paraffin. As can be seen, the DSC curves almost overlapped, which corroborates the thermal reproducibility of paraffin. The melting temperature and melting enthalpy of pure paraffin were found to be $46.92 \pm 0.53\text{ }^{\circ}\text{C}$ and $161.25 \pm 0.38\text{ J/g}$, which are almost consistent with the values reported in literature.

Fig. 8 shows the DSC curves of paraffin containing 0.25, 0.5 and 1 % GO nanoparticles. The melting temperature of the samples containing 0.25, 0.5 and 1 % GO nanoparticles was 46.17 ± 0.63 , 46.67 ± 0.15 and $45.95 \pm 2.03\text{ }^{\circ}\text{C}$, respectively. According to the obtained values, the melting temperature remained almost unchanged with the addition of GO nanoparticles to paraffin. The slight reduction (p -value > 0.05) of melting temperature in nanocomposites compared to pure paraffin can be attributed to the higher thermal conductivity of GO nanoparticles. The melting enthalpy of samples containing 0.25, 0.5 and 1 % GO nanoparticles was 151.08 ± 3.25 , 147.00 ± 4.81 and $144.83 \pm 1.48\text{ J/g}$, respectively. As can be seen, samples containing unmodified GO possess a lower enthalpy value than pure paraffin (p -value < 0.05), which

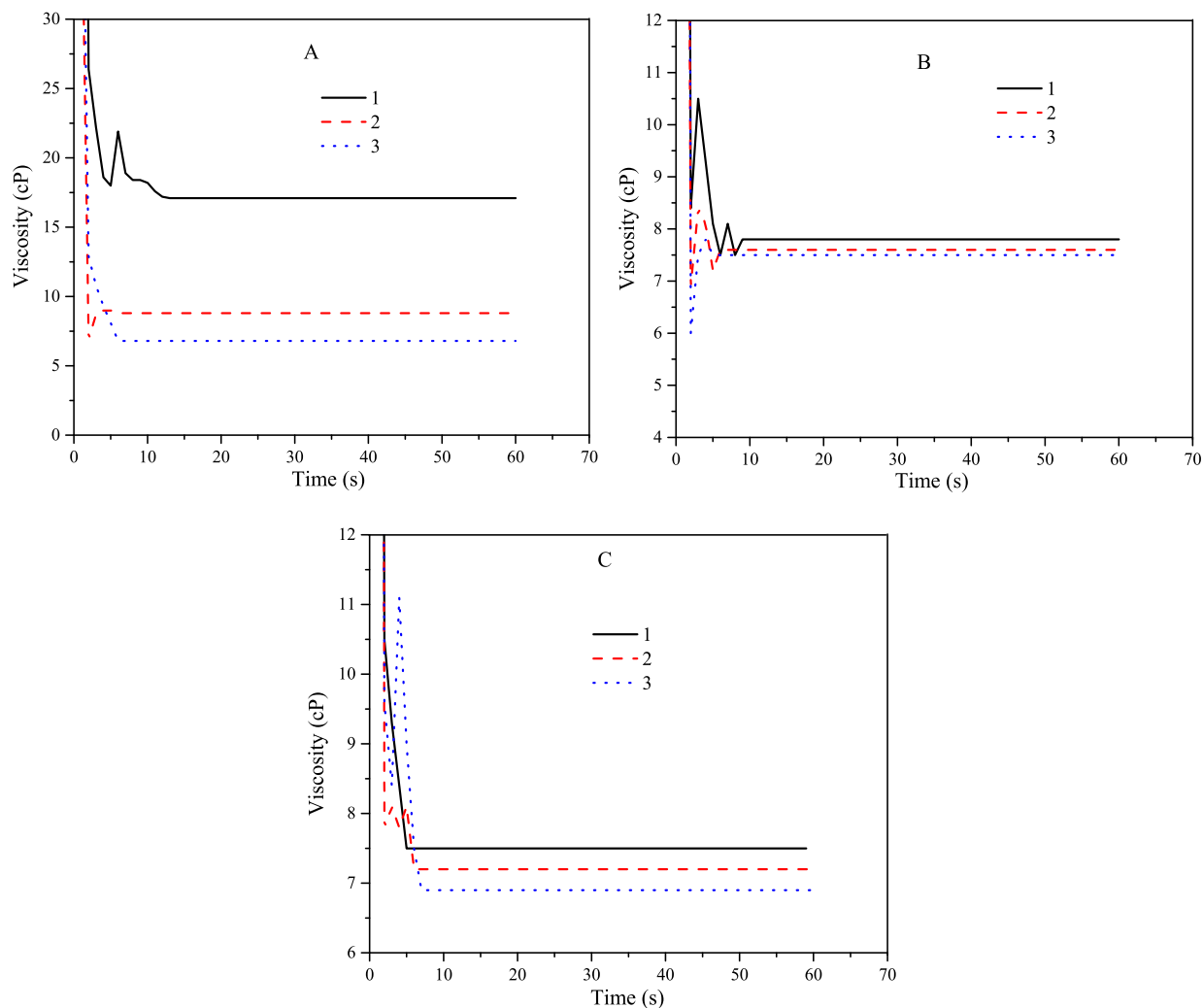


Fig. 13. P/0.5 GO nanocomposites viscosity diagram. A) $T = 40\text{ }^{\circ}\text{C}$, B) $T = 45\text{ }^{\circ}\text{C}$, C) $T = 50\text{ }^{\circ}\text{C}$.

is due to the interfering effect of nanoparticles on paraffin crystallization. The higher the amount of GO, the lower the enthalpy. Reduction of melting enthalpy by incorporating nanoparticles into PCM has been reported by other researchers. He et al. [23] confirmed that the inclusion of graphene nanoparticle in paraffin wax decreases the melting enthalpy, and this decrease was attributed to the van der Waals interactions between paraffin and graphene nanoparticle as well as the inhibitory effect of nanoparticles in heat release. Similarly, Li and Wang [24] added different amounts of amino azobenzene-modified graphene nanoparticles including 5, 10, 1nd 15 % to polyethylene glycol (PEG) and confirmed that the melting enthalpy decreases proportionally with the amount nanoparticles. The decrease of melting enthalpy was mainly attributed to the decrease in PEG fraction.

Fig. 9 illustrates the DSC curves of paraffin samples containing 0.25, 0.5, and 1 % modified GO. The melting temperature of samples comprising 0.25, 0.5, and 1 % modified GO nanoparticles was 47.52 ± 0.26 , 46.02 ± 0.38 , and $46.47 \pm 0.64\text{ }^{\circ}\text{C}$, respectively. Besides, the enthalpy of theses samples was 174.90 ± 1.72 , 161.01 ± 1.18 , and $171.47 \pm 3.46\text{ J/g}$, respectively. As can be seen, the modified nanoparticles, did not cause a decrease in enthalpy. Instead, a moderate increase in enthalpy was observed for samples containing 25 % and 1 % MGO ($p\text{-value} < 0.05$). The increase in melting enthalpy with the addition of surface modified nanoparticles has been verified by other research groups. Negar et al. [26] corroborated that the inclusion of carboxyl modified graphene nanoparticles in paraffin increases the melting enthalpy. In their work, different amount of COOH-modified

nanoparticle including 0.25, 0.5, and 0.75, and 1 % were added to paraffin and it was observed that melting enthalpy increases proportionally with the amount of nanoparticles.

In addition, in samples with modified GO, DSC curves of four repetitions almost overlapped and the thermal stability and reproducibility were higher than the samples with unmodified GO. The higher enthalpy and the better reproducibility of the samples with modified nanoparticles can be elucidated by the creation of silane groups on the surface of nanoparticles. The presence of these groups on GO surface makes the structure more organic and more compatible with paraffin.

Crystallinity of the samples was calculated by normalizing the melting enthalpy of the sample to melting enthalpy of the 100 % crystalline paraffin using Eq. (4):

$$\text{Crystallinity}\% = \frac{\Delta H_m}{\Delta H_m^0 (1-x)} \times 100 \quad (4)$$

where ΔH_m is melting enthalpy obtained from DSC curve, ΔH_m^0 is melting enthalpy of 100 % crystalline paraffin, and x is weigh fraction of nanoparticles. The crystallinity of the nanocomposites with 0.25, 0.5, and 1 wt% GO nanoparticles was found to be 89.0 ± 1.9 , 84.0 ± 2.7 , and $83 \pm 0.8\%$, respectively. Besides, the crystallinity of the nanocomposites with 0.25, 0.5, and 1 wt% MGO nanoparticles was found to be 100.0 ± 0.9 , 92.0 ± 6.8 , and $98 \pm 2.0\%$, respectively.

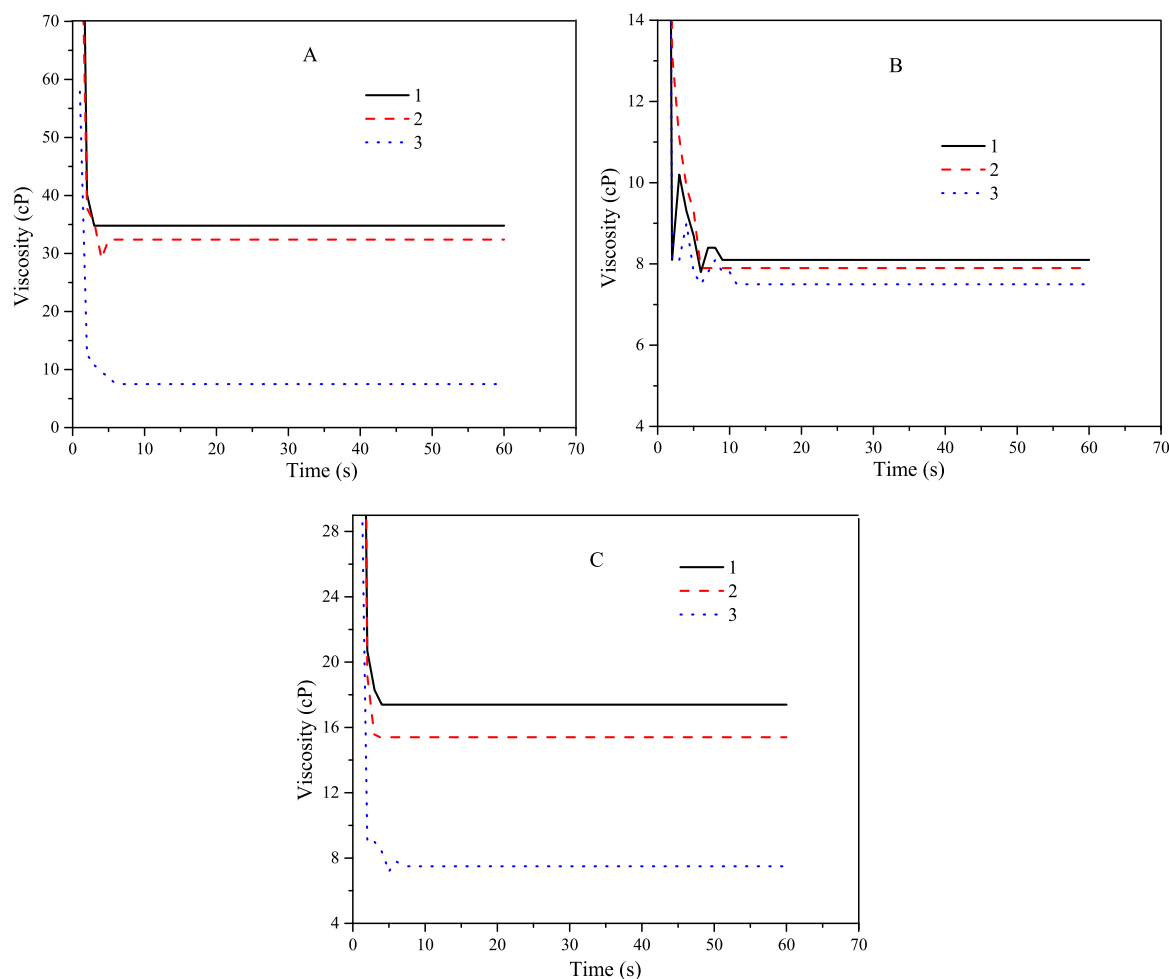


Fig. 14. P/1 GO nanocomposites viscosity diagram. A) $T = 40\text{ }^{\circ}\text{C}$, B) $T = 45\text{ }^{\circ}\text{C}$, C) $T = 50\text{ }^{\circ}\text{C}$.

3.4. TGA results

Fig. 10 displays the TGA curves of paraffin and nanocomposites with 0.25 and 1 % of pure and modified GO nanoparticles. Besides, the $T_{5\%}$, $T_{50\%}$, and $T_{90\%}$, i.e. the temperature at which 5, 20, and 90 % weight loss occurs, respectively, are collected in Table 3. As can be seen in the Fig. 10, the pure paraffin represented a one-step weigh loss in the range of 150 to 250 $^{\circ}\text{C}$ corresponding to decomposition of hydrocarbon chain. The trend of weight loss in nanocomposites was similar to pure paraffin.

According to Fig. 10 and Table 3, nanocomposite represented better thermal stability than paraffin. $T_{5\%}$ of pure paraffin was 153.34 $^{\circ}\text{C}$. $T_{5\%}$ increased to 170.27 and 166.57 $^{\circ}\text{C}$ in nanocomposites with 0.25 and 1 % GO, respectively. Moreover, $T_{5\%}$ increased to 175.35 and 178.86 $^{\circ}\text{C}$ in nanocomposites with 0.25 and 1 % MGO, respectively. $T_{50\%}$ of pure paraffin was 227.66 $^{\circ}\text{C}$. $T_{50\%}$ increased to 239.59 and 234.77 $^{\circ}\text{C}$ in nanocomposites with 0.25 and 1 % GO, respectively. Moreover, $T_{50\%}$ increased to 241.76 $^{\circ}\text{C}$ and 245.44 in nanocomposites with 0.25 and 1 % MGO, respectively. $T_{90\%}$ of pure paraffin was 251.22 $^{\circ}\text{C}$. $T_{90\%}$ increased to 262.17 and 258.65 in nanocomposites with 0.25 and 1 % GO, respectively. Moreover, $T_{90\%}$ increased to 265.15 and 271.68 $^{\circ}\text{C}$ in nanocomposites with 0.25 and 1 % MGO, respectively. The higher thermal stability of nanocomposites can be elucidated by the role of graphene sheets in preventing the release of degradation products from the sample.

Moreover, it can be observed that modified GO nanoparticles impart better thermal stability to the samples compared to the unmodified counterparts. The better thermal stability of nanocomposite loaded with ETMS-modified nanoparticles can be mainly explained by the higher

temperature required to break down the C-O-Si bond [30,41,42].

3.5. Thermal conductivity test

Table 4 summarizes the thermal conductivity values of nanocomposite containing 0.25 % unmodified and modified GO nanoparticles at two different temperatures of 20 and 45 $^{\circ}\text{C}$, i.e. in solid and liquid states. Thermal conductivity is one of the most important criteria for determining PCM performance. Higher thermal conductivity enhances the rate of heat loading and release in the PCM.

According to the literature the thermal conductivity of paraffin in liquid and solid states is about 0.10 and 0.20 W/mK, respectively [43,44]. When GO or MGO nanoparticles were added to paraffin, the thermal conductivity increased. In the liquid state, when 0.25 GO and 0.25 MGO were added to paraffin, the thermal conductivity increased to 0.139 and 0.135 W/mK, respectively. In the solid state, when 0.25 GO and 0.25 MGO were added to paraffin, the thermal conductivity increased to 0.251 and 0.226 W/mK, respectively. In other words, in liquid state, with the addition of 0.25 GO and MGO, the thermal conductivity increased by 39 and 35 %, respectively, and in solid state with the addition of 0.25 GO and MGO, the thermal conductivity increased by 25 and 13 %, respectively.

In general, when GO nanoparticles were loaded into paraffin, the thermal conductivity coefficient was raised compared to pure paraffin. The increase in thermal conductivity of paraffin with the inclusion of carbon-based nanoparticles has been confirmed by other researches. In this regard, Abu-Hamdeh et al. [45] stated that by adding 1 vol% graphene and graphene oxide to paraffin, the thermal conductivity

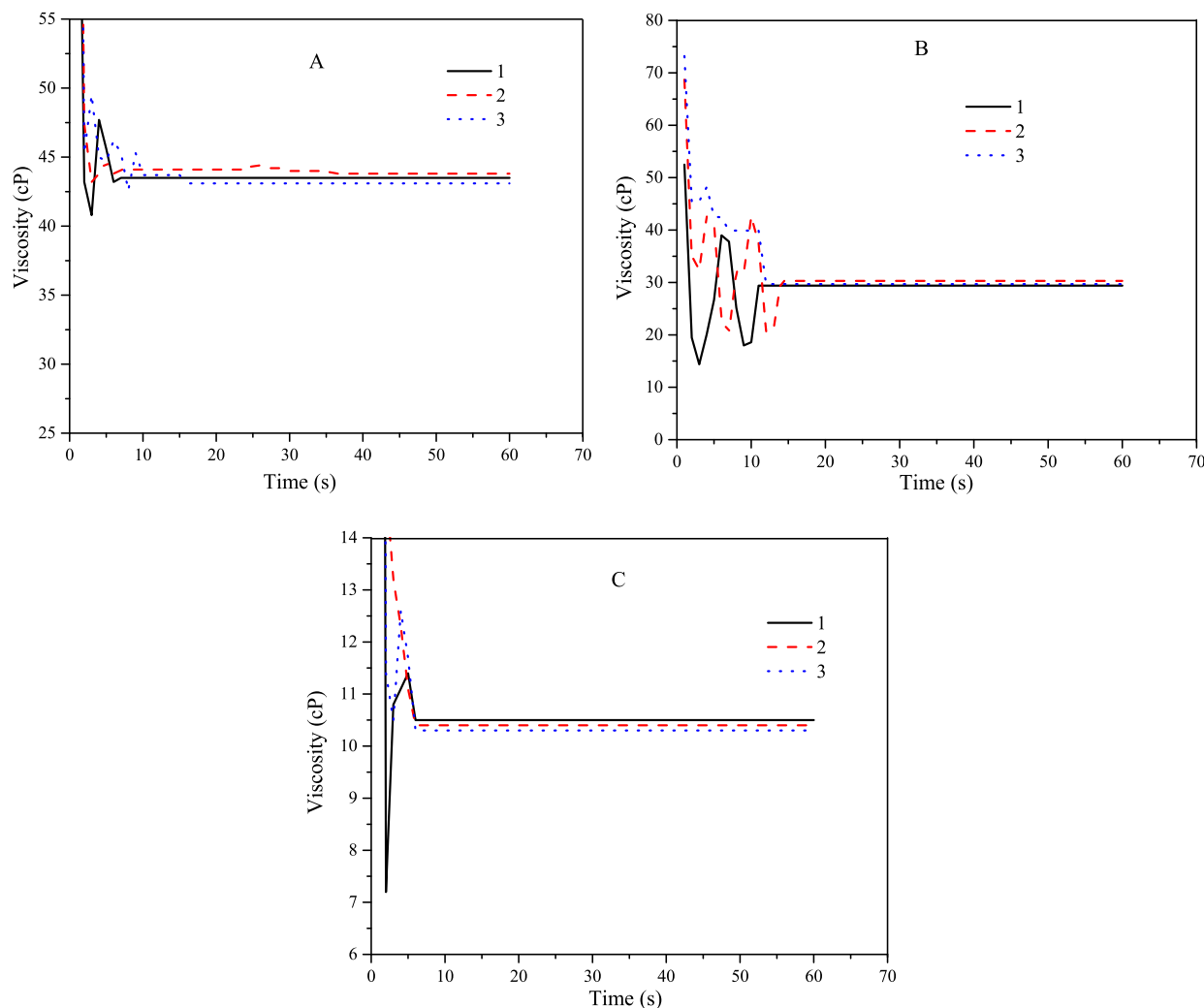


Fig. 15. P/0.25 MGO nanocomposites viscosity diagram. A) $T = 40\text{ }^{\circ}\text{C}$, B) $T = 45\text{ }^{\circ}\text{C}$, C) $T = 50\text{ }^{\circ}\text{C}$.

increased to 0.66 and 0.59 W/m.K, respectively. Similarly, Liu et al. [45] stated that when 4 wt% graphene was added to the matrix of paraffin and olefin copolymer, the thermal conductivity increased to 1 W/m.K.

According to Table 4, the thermal conductivity coefficients of the samples in the liquid state were lower than those in the solid state ($p\text{-value} < 0.05$), which can be explained by the destruction of regular crystalline structure and its transformation into a disordered liquid phase [46]. Moreover, the thermal conductivity values of nanocomposites with modified GO were almost lower than those with unmodified GO. However, the difference was not statistically significant ($p\text{-value} > 0.05$). In the solid state, thermal conductivity in the nanocomposites with MGO decreased by 10 % compared to ones with GO ($p\text{-value} = 0.07$), and in the liquid state, thermal conductivity in nanocomposites with MGO decreased by about 3 % compared to ones with GO ($p\text{-value} = 0.97$).

With surface modification of nanoparticles, their dispersion in paraffin matrix and their interaction with paraffin chains are improved. When the nanoparticles are better dispersed in the matrix, their distance becomes longer and the possibility of forming a continuous path for heat transfer is reduced, which leads to lower thermal conductivity of the composite. On the other hand, when the interaction between nanoparticles and matrix is improved, the interfacial thermal transport is boosted, which results in higher thermal conductivity. In nanocomposites with surface modified GO, these effects competed with each other and the former effect prevailed, leading to a decrease in thermal conductivity compared to unmodified GO.

If more nanoparticles are included in the paraffin matrix, the thermal conductivity is improved by surface modification, because in this case, the distance between the nanoparticles that are uniformly dispersed in the matrix is reduced and a continuous path for heat transfer is developed in the composite. In this case, the heat transfer is improved with better interfacial compatibility of nanoparticles and matrix. Therefore, it is anticipated that in high-amount of nanoparticles, surface modification can have a positive effect and increase the thermal conductivity [46].

However, the stability and reproducibility of the thermal conductivity values of nanocomposites with modified GO were significantly higher than those with unmodified GO. It should be noted that PCMs should be practical and stable. They go through many thermal cycles and should be able to retain their properties at the end of the cycles. Therefore, although the thermal conductivity values of the composites with MGO are slightly lower than those with unmodified GO, the MGO is a better choice for developing a practical PCM due to having more reproducible thermal behavior.

3.6. Viscosity measurement analysis

Viscosity of paraffin and nanocomposites was measured at different temperatures of 40, 45, and 50 $^{\circ}\text{C}$. Each measurement was performed in triplicate. Viscosity changes versus time are represented in Figs. 11–17. As shown in Fig. 11, in pure paraffin, the viscosity decreased with increasing temperature and remained almost constant after repeating the experiment.

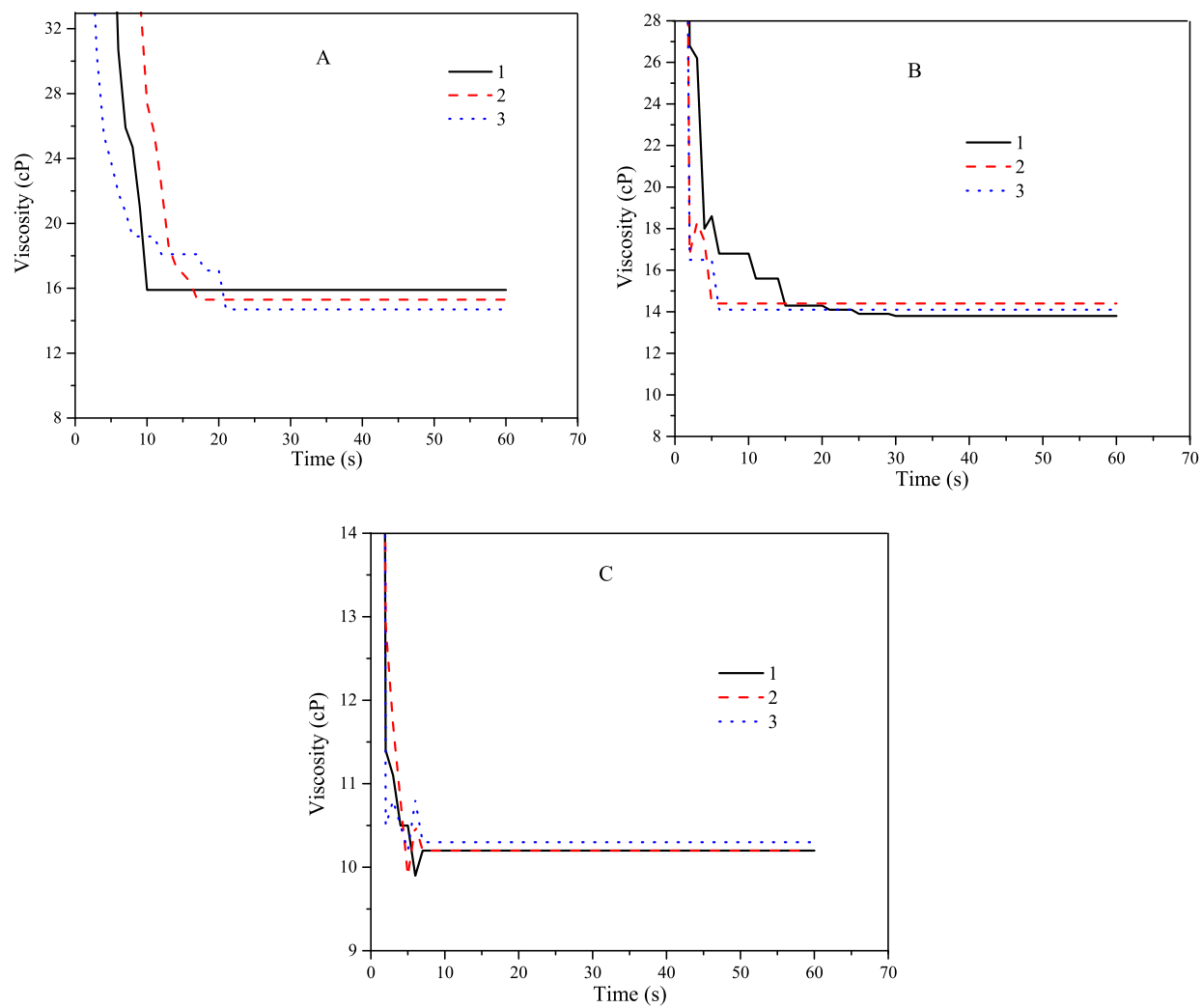


Fig. 16. P/0.5 MGO nanocomposites viscosity diagram. A) T = 40 °C, B) T = 45 °C, C) T = 50 °C.

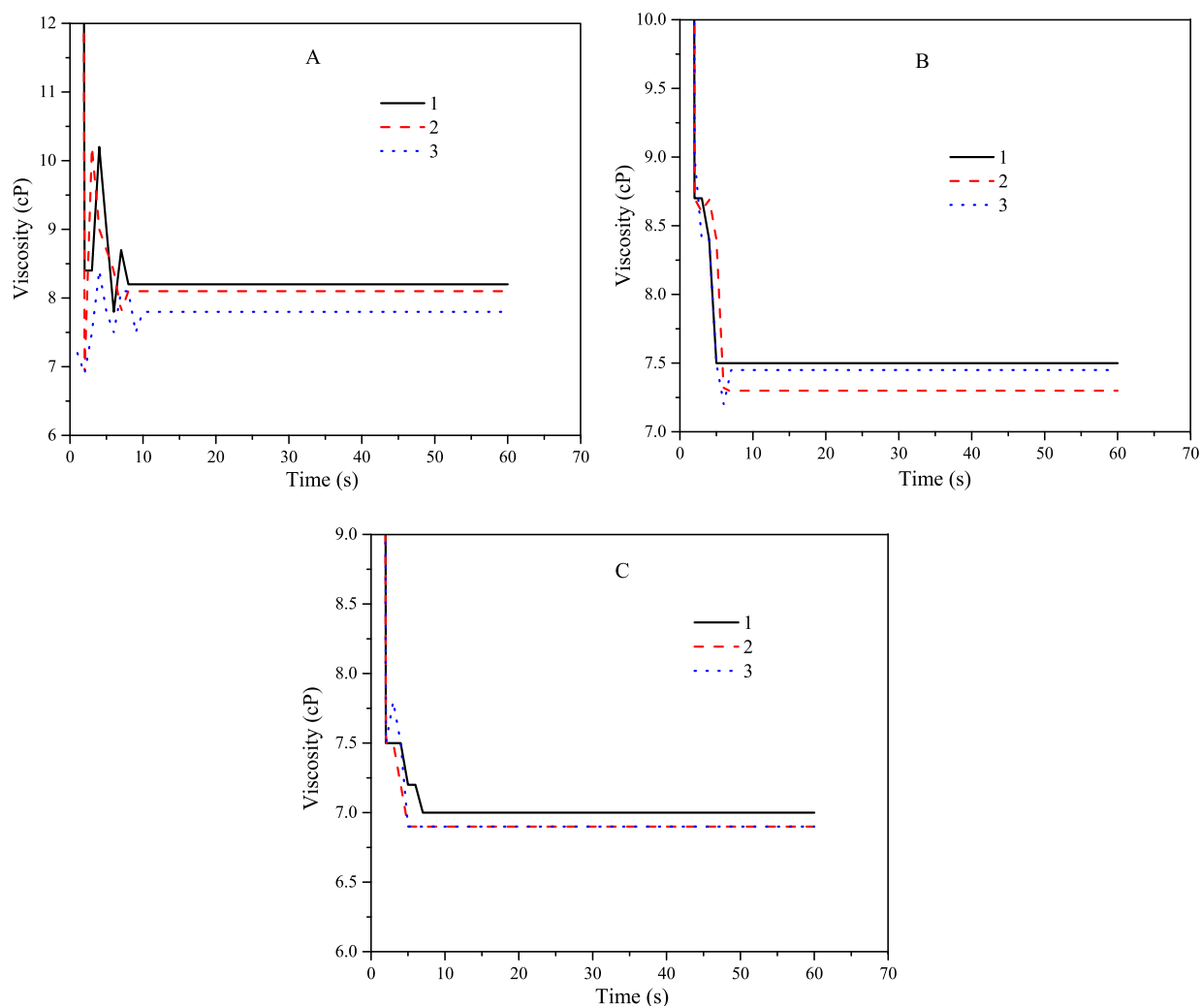


Fig. 17. P/1 MGO nanocomposites viscosity diagram. A) T = 40 °C, B) T = 45 °C, C) T = 50 °C.

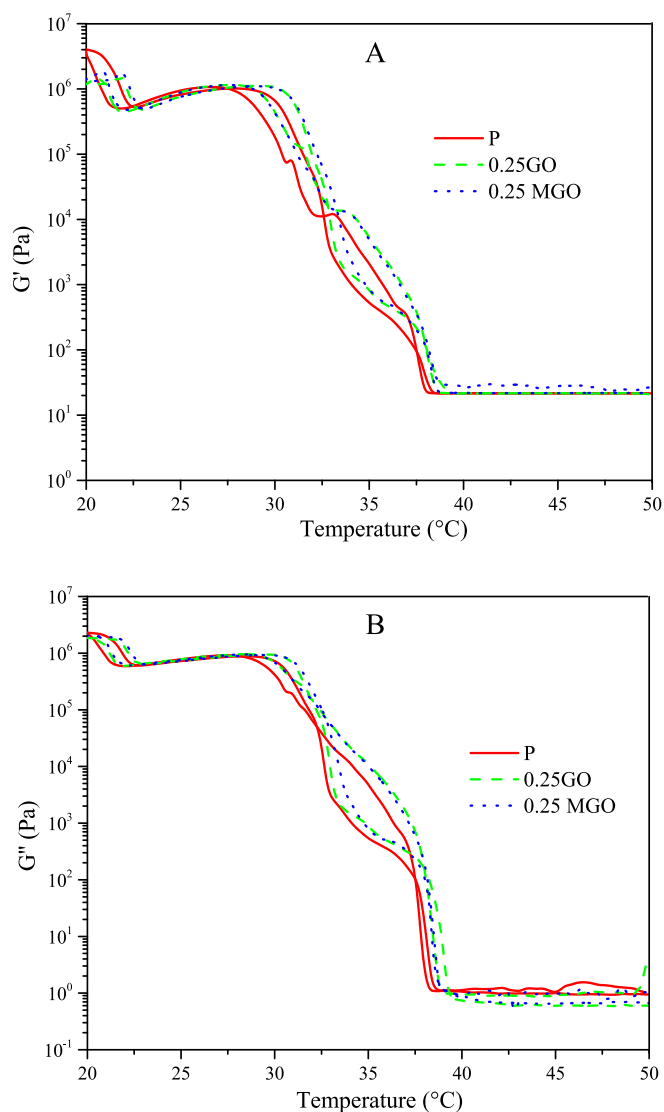


Fig. 18. Storage modulus (A), and loss modulus (B) versus temperature pure paraffin and nanocomposites with 0.25 % GO and MGO.

The viscosity of nanocomposites with unmodified GO nanoparticles was higher than the pristine paraffin, which is most likely due the confinement and interaction effects of nanoparticles on paraffin chains. Regarding the first effect, GO nanoparticles impede the movement of paraffin chains and reduce chain dynamics. Regarding the second effect, van der Waals interactions between GO and hydrocarbon chains increase the viscosity. Moreover, in nanocomposites with unmodified GO, when the experiment was repeated, the viscosity was reduced, which could be explained by the weak interaction forces between paraffin and unmodified nanoparticles and the sedimentation of nanoparticles.

The viscosity of nanocomposites with modified GO was much higher than pristine paraffin and nanocomposites with unmodified GO. Because as a result of surface modification, the interaction between GO nanoparticles and paraffin chains was greatly promoted. Moreover, in nanocomposites with modified GO, the viscosity did not decrease after repeating the experiment. Because, the existence of a great number of interactions between MGO nanoparticles and paraffin chains prevented the sedimentation of nanoparticles. In fact, by using the modified nanoparticles, viscosity stability was increased.

In addition, according to the obtained results, the viscosity did not increase proportionally to the amount of GO. With the increase of nanoparticles, the viscosity first rose and then fell. The composition with

0.25 % nanoparticles (GO and MGO) represented the highest viscosity, indicating a better dispersion of nanoparticles in this percentage. When higher amount of GO was added to the composition, the nanoparticles were agglomerated and the available surface area for the interaction between them and the paraffin chains was reduced. More precisely, at 0.25 %, there is the best relationship between the surface of nanoparticles and paraffin, and increasing the percentage of nanoparticles decreases this relationship.

3.7. Rheology analysis

In the rheology test, the changes of storage modulus (G') and the loss modulus (G'') against temperature are plotted. This test was conducted of pure paraffin and nanocomposites with 0.25 % GO and MGO. As can be seen in Fig. 18, the storage and loss moduli of nanocomposites are higher than those of pure paraffin, highlighting the reinforcing effect of GO on paraffin [47,48]. Moreover, the storage and loss moduli of nanocomposites with MGO were higher than those of nanocomposites with GO, confirming the higher interaction between modified nanoparticles and paraffin.

3.8. Melt fraction analysis

The curves of melting fraction versus time for pure paraffin and nanocomposites at different temperatures of 40, 45, and 50 °C are shown in Figs. 19–21. The melting fraction of PCM is determined by the thermal conductivity and latent heat of the samples. According to the obtained graphs (Figs. 19–21), nanocomposites demonstrate a decrease in melting time compared to paraffin, most likely due to the improvement of thermal conductivity and reduction of latent heat of melting. The time required for complete melting at 40 °C for pure paraffin was about 120 min, and in nanocomposites with 0.25, 0.5, and 1 % GO, this time decreased to 92, 104, and 108 min, respectively. This time at 45 °C for pure paraffin was about 42 min, and in nanocomposites with 0.25, 0.5, and 1 % GO, this time decreased to 37, 39, and 39 min, respectively. And finally the time needed for complete melting at 50 °C for pure paraffin was about 28 min, and in nanocomposites with 0.25, 0.5, and 1 % GO, this time decreased to 21, 22, and 23 min, respectively. The time required for complete melting was reduced by 12–25 % in nanocomposite with 0.25 % GO compared to the pristine paraffin.

According to the obtained results, the greatest decrease in complete melting time was observed for the sample with 25 % nanoparticles, which again indicates the optimum properties in this composition. Moreover as can be seen, with increasing temperature, the time difference for complete melting decreases, showing that the temperature potential has dominated the thermal conductivity difference potential [35].

4. Conclusion

Surface modification of nanoparticles is a practical approach to improve the thermal properties and stability as well as the heat storage performance of composite phase change materials (PCMs). In the present work, for the first time, graphene oxide nanoparticles were modified with a silane component and used for the preparation of paraffin-based composite PCM. Different amount of modified and unmodified GO nanoparticles including 0.25, 0.5, and 1 wt% were added to paraffin and the properties of the resulting nanocomposites were evaluated by many techniques. FTIR spectrum confirmed the successful formation of silane component on the GO surface. XRD pattern revealed the main crystalline peak at $2\theta = 12.16^\circ$ with d-spacing of 0.73 nm and crystallite size of 3.77 nm. AFM image showed that the modified nanoparticles have a thickness of 1–2 nm, slightly more than the thickness of a single-layer GO. DSC thermograms disclosed that melting enthalpy decreases by 6–10 % in nanocomposites with unmodified GO, while it increases by 0–8 % in nanocomposites with modified GO compared to pure paraffin.

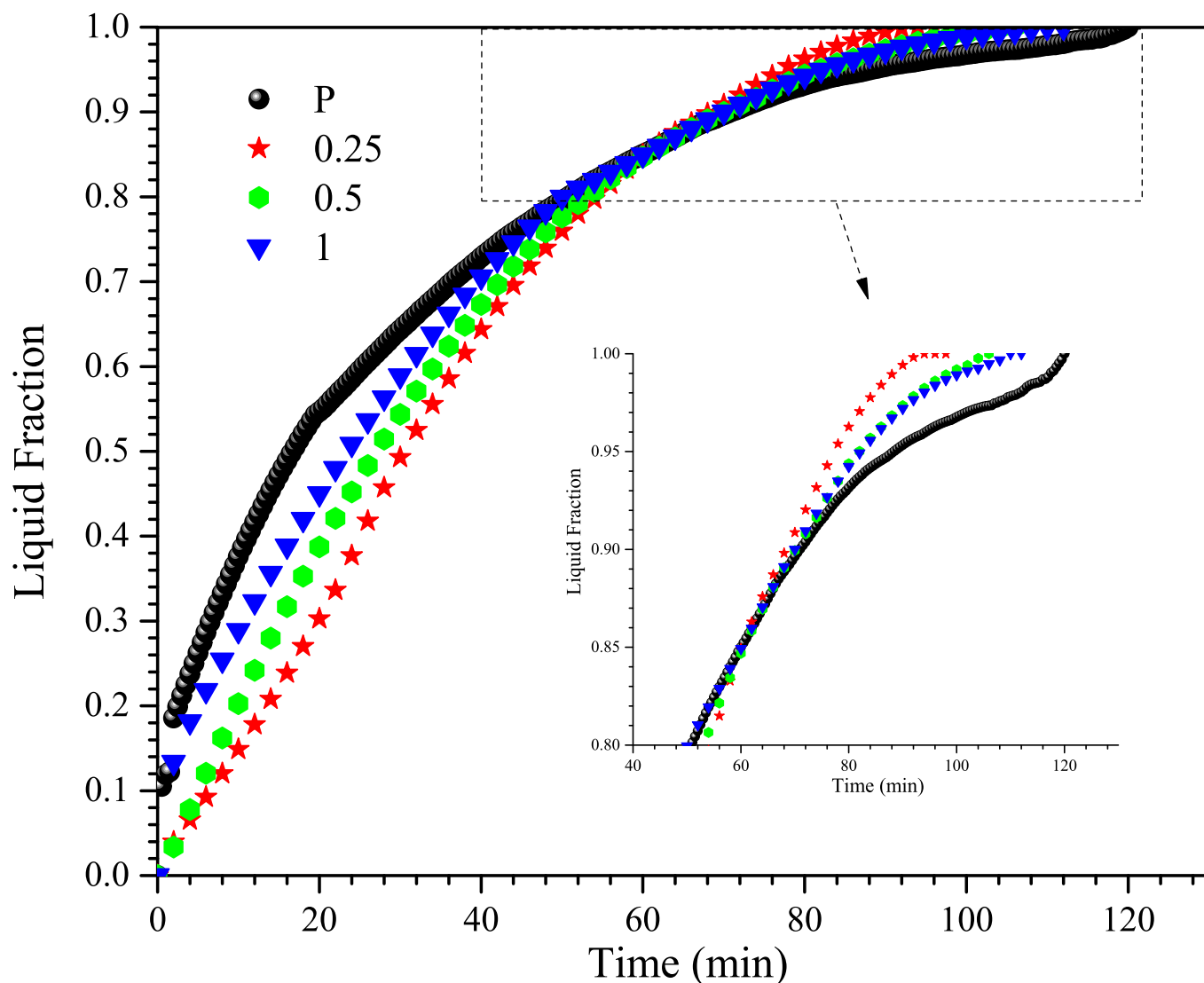


Fig. 19. Liquid fraction PCM analysis at 40 °C.

TGA curves indicated that initial degradation temperature increases about 13–17 °C and 22–26 °C for nanocomposites with unmodified and modified GO, respectively. Thermal conductivity measurement showed that the thermal conductivity of nanocomposites is higher than that of pure paraffin and the thermal reproducibility of the modified sample is higher than that of the unmodified and pure samples. Viscosity measurement substantiated that the surface modification of nanoparticles prevents the sedimentation of nanoparticle and improves the viscosity stability. Melting fraction evaluation verified that the time required for complete melting in the nanocomposite with 0.25 % GO is reduced by 12–25 % compared to the pristine paraffin. To sum up, the nanocomposite comprising 0.25 % modified GO represented superior thermal properties and stability and can be considered as a good choice for developing a practical phase change material.

Research ethics

We further confirm that any aspect of the work covered in this manuscript that has involved human patients has been conducted with the ethical approval of all relevant bodies and that such approvals are acknowledged within the manuscript.

IRB approval was obtained (required for studies and series of 3 or more cases).

Written consent to publish potentially identifying information, such as details or the case and photographs, was obtained from the patient(s) or their legal guardian(s).

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Author agreement statement

We the undersigned declare that this manuscript is original, has not been published before and is not currently being considered for publication elsewhere.

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

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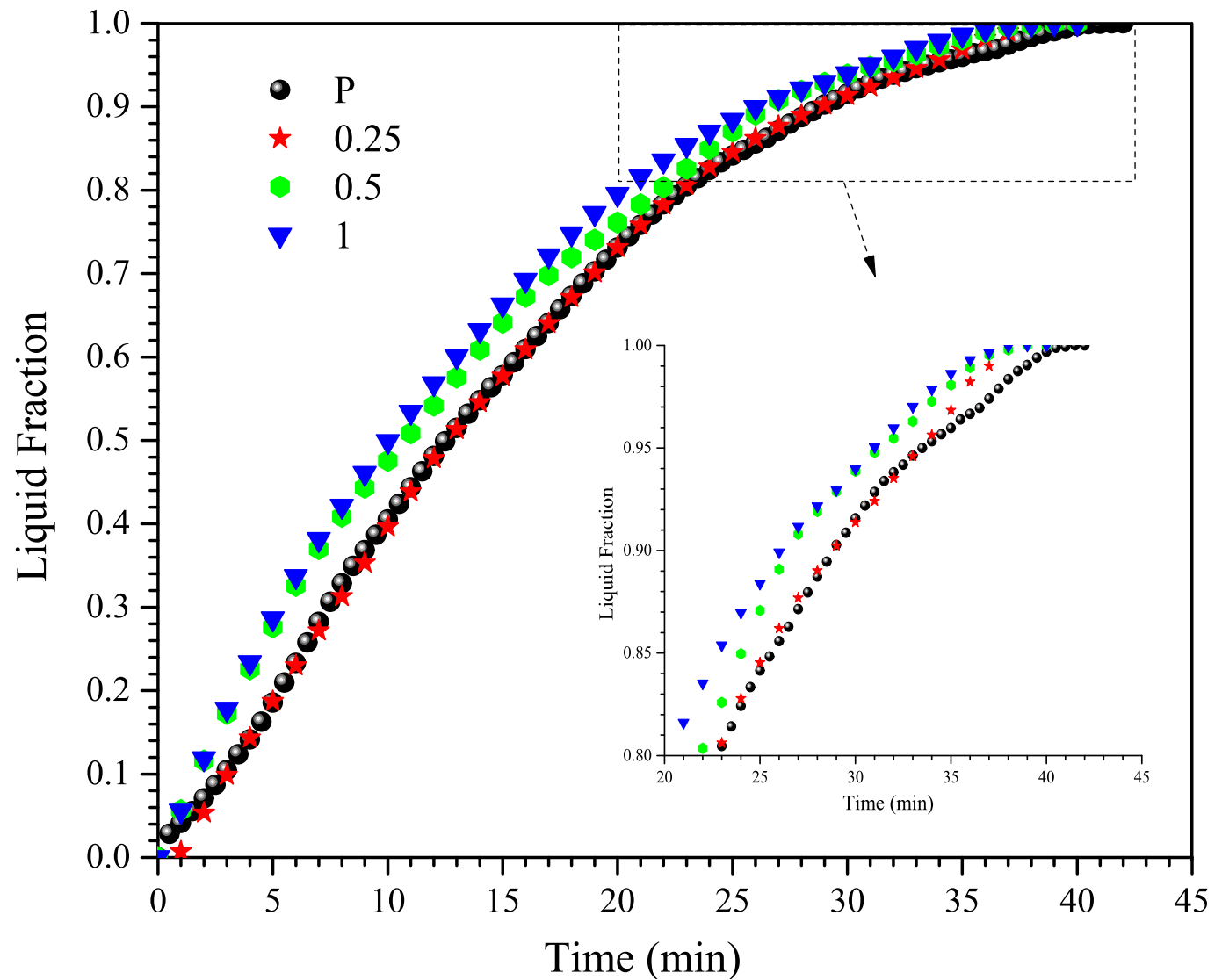


Fig. 20. Liquid fraction PCM analysis at 45 °C.

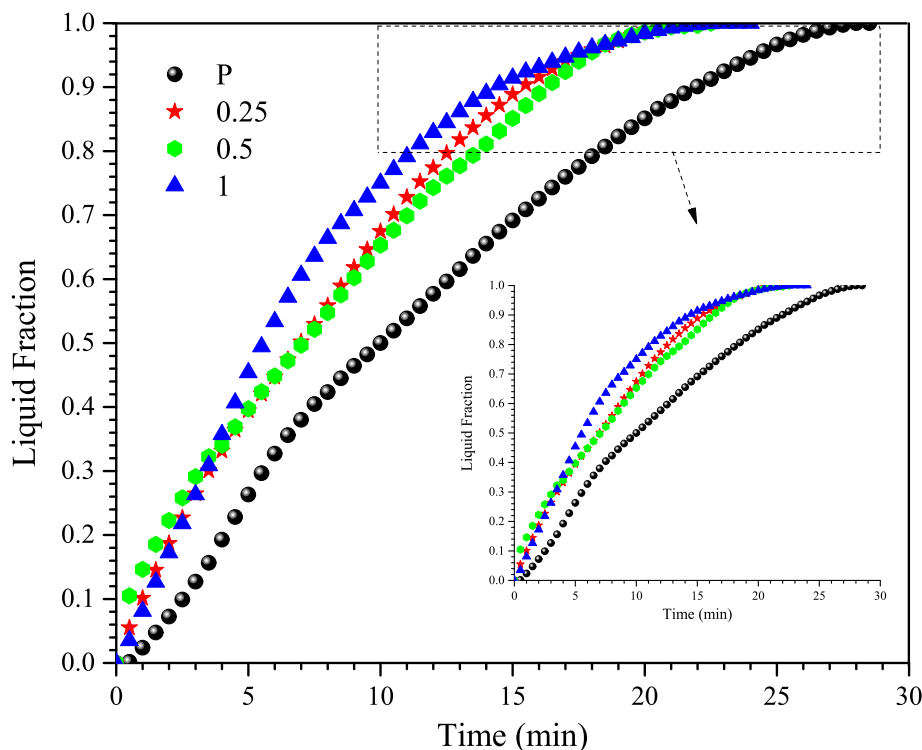


Fig. 21. Liquid fraction PCM analysis at 50 °C.

CRediT authorship contribution statement

Mehdi Ghaffari: Writing – review & editing, Supervision, Project administration, Investigation. **Seyed Mohammad Javad Hosseini Khsari:** Methodology. **Rasool Bahrapoury:** Writing – original draft, Investigation.

Declaration of competing interest

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

Data availability

No data was used for the research described in the article.

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